

Parent Guide — senior / module-09- subtraction-nikhilam

IKS Curriculum

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Parent Guide — Subtraction — Nikhilam (All from 9) (Senior)

What is your child learning?

Nikhilam Navataścaramam Daśataḥ — “all from 9, the last from 10” — is Tirthaji’s second sūtra. It produces fast complements to powers of 10, removing the need for borrowing in many subtractions. Conceptually identical to two’s-complement binary arithmetic used in modern computers.

Why this matters

This module is part of an Indian Knowledge Systems curriculum that introduces classical Indian frameworks alongside their modern-scientific parallels. Your child will learn to (a) name and use precise Sanskrit terms, (b) understand the observational basis for the classical framework, and (c) relate it to current scientific or scholarly understanding.

What you can do at home

Your child is learning a fast mental subtraction method. Ask them to demonstrate by computing change at the shop, or how much time until dinner. The goal is fluent number sense.

Conversation prompts

Try one of these over a meal or walk:

1. “What’s one Sanskrit word you learned this week? What does it mean?”
2. “What did you find surprising in this module?”
3. “Is there anything you’re not sure about? Let’s look it up together.”
4. “How would you explain this to your grandmother / grandfather?”

Questions parents often ask

Q: Is this religious instruction? A: No. The module presents Indian classical frameworks (cosmology, mathematics, astronomy, ecology) as historical knowledge systems with their own logic and method. The framing is scholarly and comparative, not devotional.

Q: Will my child be tested on Sanskrit? A: Sanskrit terms are introduced for precision, not for rote memorisation. The summative quiz asks students to *use* the terms — for example, “Which doṣa is dominant in the morning hours?” — rather than translate them.

Q: How does this connect to NCERT / board syllabus? A: This module supplements (does not replace) NCERT content. NEP 2020 §4.6 calls for the integration of Indian Knowledge Systems. The pack’s README.md lists the specific NEP principles each module advances.

Reading further (for parents)

See sources.md in this pack for the full source list.

10-Day Lesson Plan

Lesson Plan — Subtraction — Nikhilaṃ (All from 9) (Senior)

10-day arc, 45 min per session.

Day 1 — The Sūtra — All from 9, Last from 10

Objective: Compute the complement of any 2-digit number to 100.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Complement to 1000 and Beyond

Objective: Compute complements to 1000 and 10000 mentally.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Subtract from a Power of 10

Objective: Compute 1000–567 and similar instantly.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — General Subtraction via Complements

Objective: Use add-the-complement trick for arbitrary subtractions.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Why It Works — The Algebra

Objective: Derive the place-value identity that makes the rule work.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Bar Model Proof

Objective: Draw a visual proof using bar models and number lines.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Two's Complement in Computers

Objective: Show how the same idea underlies binary computer subtraction.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Speed Day

Objective: Complete a 30-problem mental subtraction sprint.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Mixed Practice

Objective: Solve a mixed problem set comparing complement vs traditional borrow.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Assessment

Objective: Take the summative quiz.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Homework

Homework — Subtraction — Nikhilam (All from 9) (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sūtra — All from 9, Last from 10** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Complement to 1000 and Beyond** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Subtract from a Power of 10** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — General Subtraction via Complements** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Bar Model Proof** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Two's Complement in Computers** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Speed Day** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.